

Fondation
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Systèmes Réseaux et Télécommunications

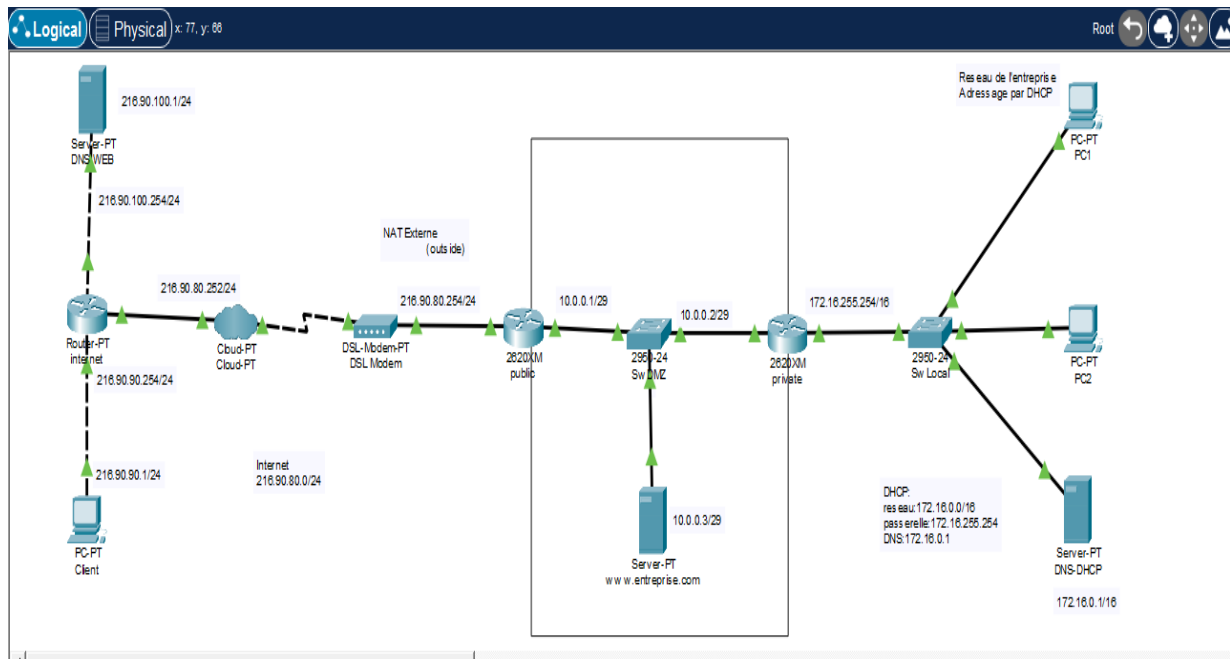
Licence 3 en Systèmes Réseaux et Télécommunications

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Compte rendu TP : ACL et DMZ

1. Le réseau : Réalisons la structure de réseau suivante



2. Routage et DNS

Appliquons à tous les routeurs un protocole dynamique

✓ *Routeur Internet*

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    10.0.0.0/8 [120/1] via 216.90.80.254, 00:00:08, FastEthernet9/0
R    172.16.0.0/16 [120/2] via 216.90.80.254, 00:00:08, FastEthernet9/0
C    216.90.80.0/24 is directly connected, FastEthernet9/0
C    216.90.90.0/24 is directly connected, FastEthernet1/0
C    216.90.100.0/24 is directly connected, FastEthernet0/0
```

✓ *Routeur public*

```

public>sh
public>show ip
public>show ip r
public>show ip ro
public>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/28 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, FastEthernet1/0
R       172.16.0.0/16 [120/1] via 10.0.0.2, 00:00:28, FastEthernet1/0
C       216.90.80.0/24 is directly connected, FastEthernet0/0
R       216.90.90.0/24 [120/1] via 216.90.80.252, 00:00:15, FastEthernet0/0
R       216.90.100.0/24 [120/1] via 216.90.80.252, 00:00:15, FastEthernet0/0

public>

```

✓ *Routeur private*

```

private>sh
private>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/29 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, FastEthernet1/0
C       172.16.0.0/16 is directly connected, FastEthernet0/0
R       216.90.80.0/24 [120/1] via 10.0.0.1, 00:00:20, FastEthernet1/0
R       216.90.90.0/24 [120/2] via 10.0.0.1, 00:00:20, FastEthernet1/0
R       216.90.100.0/24 [120/2] via 10.0.0.1, 00:00:20, FastEthernet1/0

private>

```

Tableau des résultats - Adresses traduitées

Source	Destination	Résultat - Adresses traduitées
PC1 ou 2 (172.16.x.x)	Client (216.90.90.1)	L'adresse source est traduitée en 216.90.80.254 (IP publique du routeur).
(10.0.0.3)	Client (216.90.90.1)	L'adresse source est traduitée en son IP publique statique 216.90.80.253.

Client (216.90.90.1)	PC1 ou 2 (172.16.x.x)	Échec par défaut (sauf si une session a été initiée par le PC interne) car le NAT dynamique bloque les flux entrants inconnus.
Client (216.90.90.1)	(216.90.80.253)	L'adresse de destination 216.90.80.253 est tradlatée par le routeur en adresse interne 10.0.0.3.

L'adresse renvoyée par le serveur DNS à ce stade est l'adresse IP réelle (interne) du serveur Web, soit 10.0.0.3.

Vérifions que tous les postes et serveurs puissent communiquer entre eux

✓ *Ping Client vers DNS-WEB*

```
PC>ping 216.90.100.1

Pinging 216.90.100.1 with 32 bytes of data:

Reply from 216.90.100.1: bytes=32 time=1ms TTL=127
Reply from 216.90.100.1: bytes=32 time=0ms TTL=127
Reply from 216.90.100.1: bytes=32 time=0ms TTL=127
Reply from 216.90.100.1: bytes=32 time=0ms TTL=127

Ping statistics for 216.90.100.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

✓ *Ping Client vers serveur DNS-DHCP*

```
PC>ping 172.16.0.1

Pinging 172.16.0.1 with 32 bytes of data:

Reply from 172.16.0.1: bytes=32 time=42ms TTL=125
Reply from 172.16.0.1: bytes=32 time=43ms TTL=125
Reply from 172.16.0.1: bytes=32 time=42ms TTL=125
Reply from 172.16.0.1: bytes=32 time=43ms TTL=125

Ping statistics for 172.16.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 42ms, Maximum = 43ms, Average = 42ms

PC>
```

✓ Ping Client vers serveur www.entreprise.com

```
PC>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time=54ms TTL=126
Reply from 10.0.0.3: bytes=32 time=42ms TTL=126
Reply from 10.0.0.3: bytes=32 time=53ms TTL=126
Reply from 10.0.0.3: bytes=32 time=42ms TTL=126

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 42ms, Maximum = 54ms, Average = 47ms

PC>
```

✓ Ping PC1 vers serveur DNS-WEB

```
C:\>ping 216.90.100.1

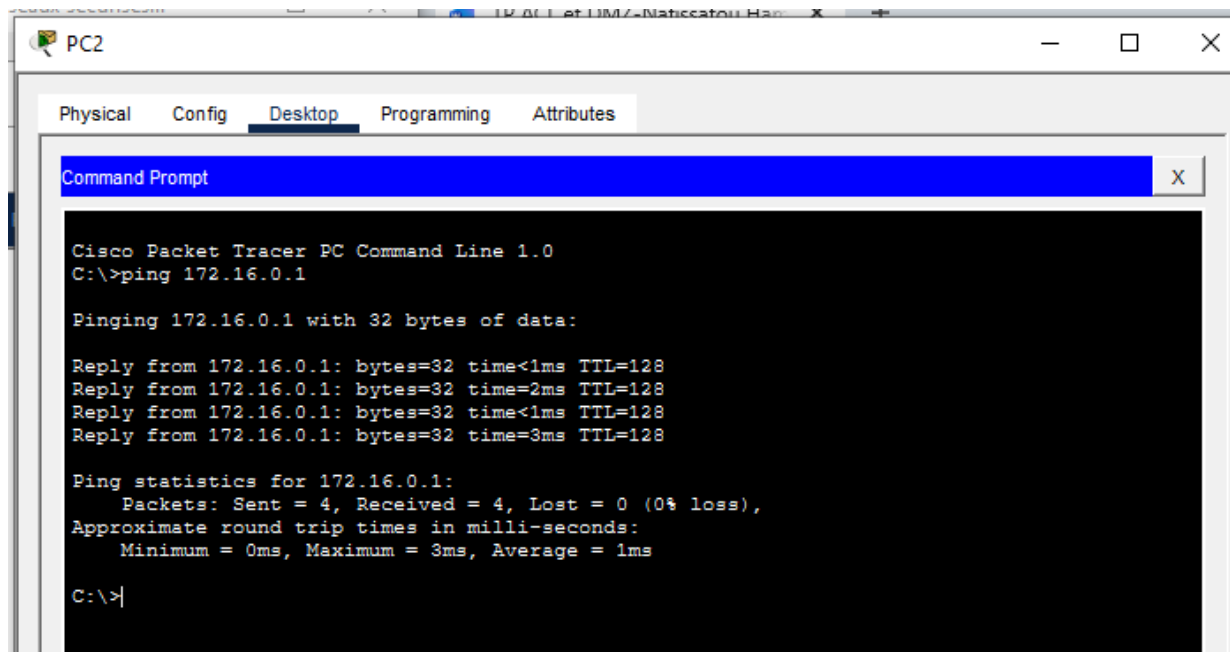
Pinging 216.90.100.1 with 32 bytes of data:

Reply from 216.90.100.1: bytes=32 time=58ms TTL=125
Reply from 216.90.100.1: bytes=32 time=63ms TTL=125
Reply from 216.90.100.1: bytes=32 time=56ms TTL=125
Reply from 216.90.100.1: bytes=32 time=57ms TTL=125

Ping statistics for 216.90.100.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 56ms, Maximum = 63ms, Average = 58ms

C:\>
```

✓ Ping PC2 vers serveur DNS-DHCP



The screenshot shows a Cisco Packet Tracer PC Command Prompt window for PC2. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a black command prompt with white text. The text shows the execution of the command 'ping 172.16.0.1', which results in four successful replies with varying round-trip times (1ms, 2ms, 1ms, 3ms) and a 0% loss rate. The statistics show 4 packets sent and received.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.0.1

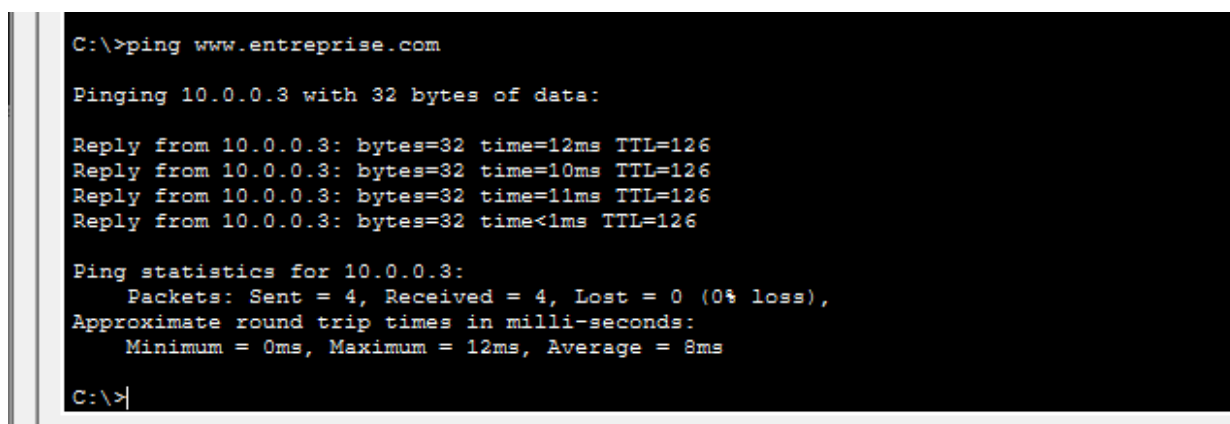
Pinging 172.16.0.1 with 32 bytes of data:

Reply from 172.16.0.1: bytes=32 time<1ms TTL=128
Reply from 172.16.0.1: bytes=32 time=2ms TTL=128
Reply from 172.16.0.1: bytes=32 time<1ms TTL=128
Reply from 172.16.0.1: bytes=32 time=3ms TTL=128

Ping statistics for 172.16.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>|
```

On constate que tous les postes et serveurs communiquent entre eux ...



The screenshot shows a Cisco Packet Tracer PC Command Prompt window for PC2. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a black command prompt with white text. The text shows the execution of the command 'ping www.entreprise.com', which results in four successful replies with varying round-trip times (12ms, 10ms, 11ms, <1ms) and a 0% loss rate. The statistics show 4 packets sent and received.

```
C:\>ping www.entreprise.com

Pinging 10.0.0.3 with 32 bytes of data:

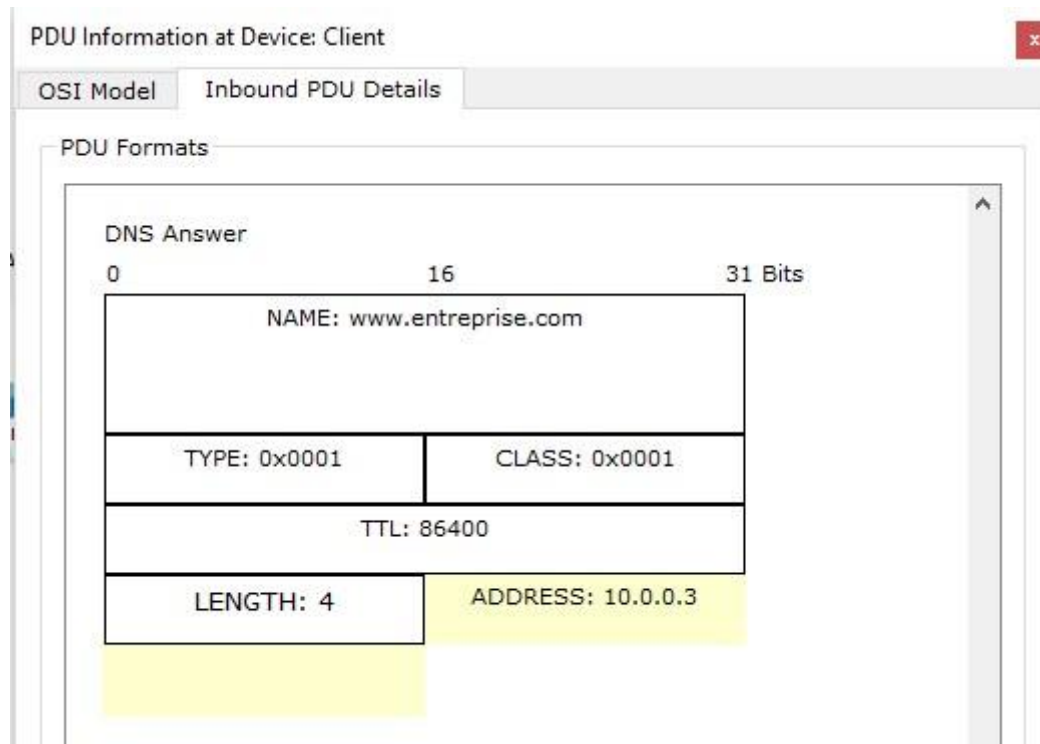
Reply from 10.0.0.3: bytes=32 time=12ms TTL=126
Reply from 10.0.0.3: bytes=32 time=10ms TTL=126
Reply from 10.0.0.3: bytes=32 time=11ms TTL=126
Reply from 10.0.0.3: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 8ms

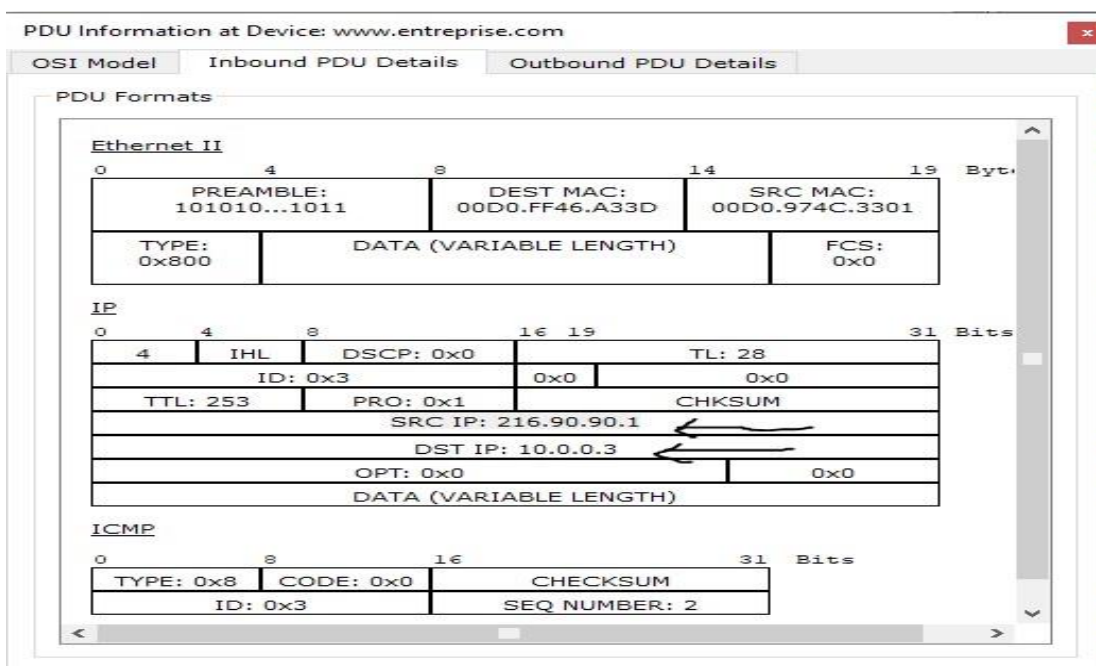
C:\>|
```

Nous parvenons à pinguer le www.entreprise.com depuis les PC1 et 2

Consultons la partie DNS de la trame L'adresse renvoyée par le DNS



Cliquons sur l'enveloppe du message final ICMP. Les adresses source et destination ?



Adresse IP source : 10.0.0.3

Adresse IP destination : 216.90.90.1

3. Translation d'adresses

Indiquons au routeur ses interfaces interne et externe

Public(config)#interface fa0/0

Private(config)#interface Eth1/0

❖ Ecrivons la règle de translation nat statique Sur le routeur public

```
public(config)#int fa0/0
public(config-if)#no ip nat outside
public(config-if)#int fal/0
public(config-if)#ip nat inside
public(config-if)#exit
public(config)#access-list 1 permit 216.90.80.0 0.0.0.255
public(config)#nat inside source list 1 int fal/0 overload
^
% Invalid input detected at '^' marker.

public(config)#ip nat inside source list 1 int fal/0 overload
public(config)#no shutdown
^
% Invalid input detected at '^' marker.

public(config)#
```

```
private(config)#int Eth1/0
%Invalid interface type and number
private(config)#int fal/0
private(config-if)#ip nat o
private(config-if)#ip nat outside
private(config-if)#exit
private(config)#
```

❖ Procédons au test de connectivite

Procédons au test de connectivite, consultons les trames pour observer la translation d'adresse

✓ PC1 vers Client


```
C:\>ping 216.90.90.1

Pinging 216.90.90.1 with 32 bytes of data:

Reply from 216.90.90.1: bytes=32 time=49ms TTL=125
Reply from 216.90.90.1: bytes=32 time=43ms TTL=125
Reply from 216.90.90.1: bytes=32 time=47ms TTL=125
Reply from 216.90.90.1: bytes=32 time=44ms TTL=125

Ping statistics for 216.90.90.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 43ms, Maximum = 49ms, Average = 45ms

C:\>
```

✓ PC2 vers Client

```
C:\>ping 216.90.90.1

Pinging 216.90.90.1 with 32 bytes of data:

Reply from 216.90.90.1: bytes=32 time=52ms TTL=125
Reply from 216.90.90.1: bytes=32 time=50ms TTL=125
Reply from 216.90.90.1: bytes=32 time=43ms TTL=125
Reply from 216.90.90.1: bytes=32 time=43ms TTL=125

Ping statistics for 216.90.90.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 43ms, Maximum = 52ms, Average = 47ms

C:\>
```

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type	Info
	0.887	--	SwDMZ	STP	
	0.887	--	PC1	ICMP	
	0.888	SwDMZ	private	STP	
	0.888	SwDMZ	www.ent...	STP	
	0.888	SwDMZ	public	STP	
	0.888	PC1	SwLocal	ICMP	
	0.889	SwLocal	private	ICMP	
	0.890	private	SwDMZ	ICMP	
	0.891	SwDMZ	public	ICMP	

Reset Simulation

☒ Constant Delay

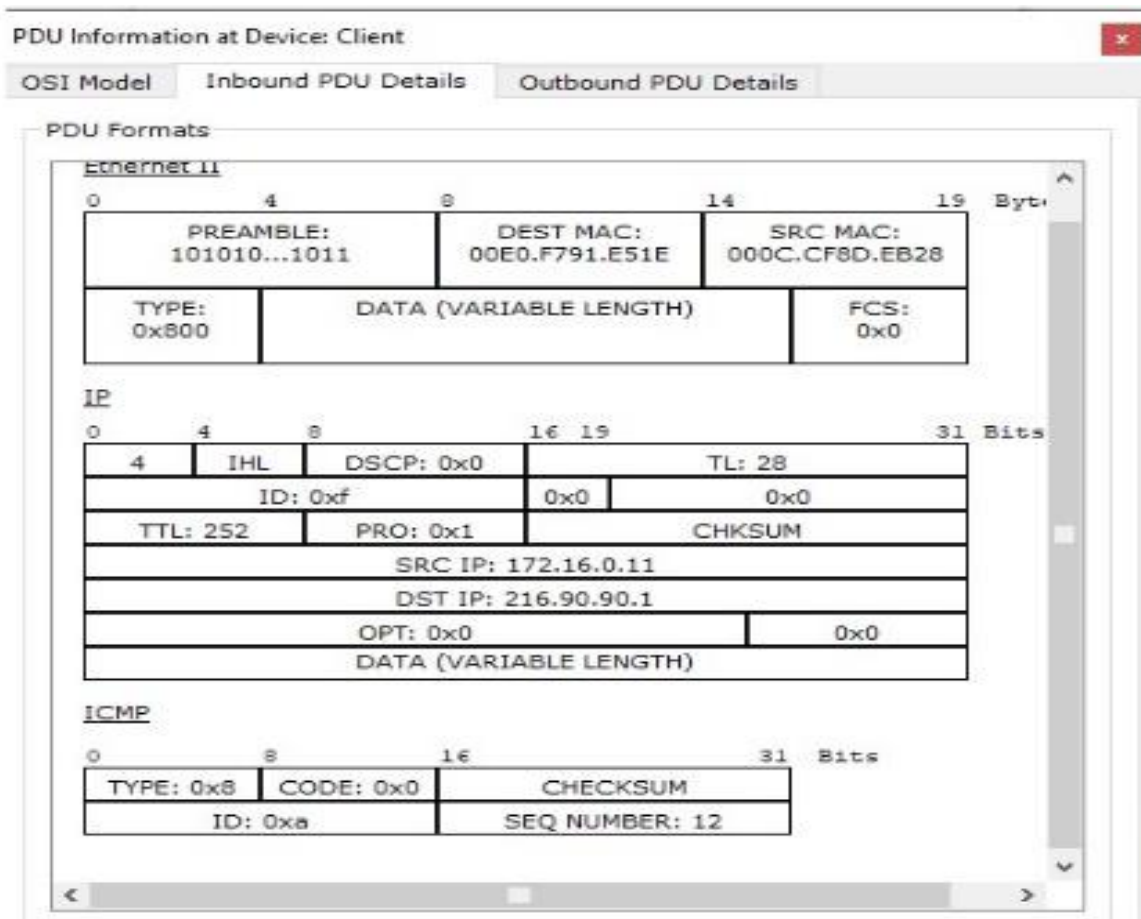
Captured to: *
0.897 s

Play Controls

Back

Auto Capture / Play

Capture / Forward



✓ www.entreprise.com vers Client

```

Packet Tracer SERVER Command Line 1/0
SERVER>ping 216.90.90.1

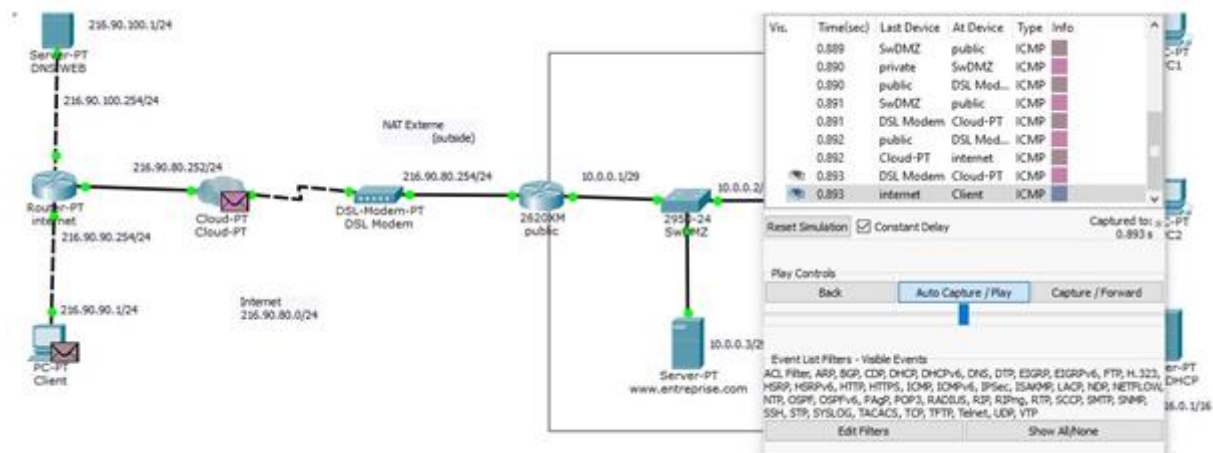
Pinging 216.90.90.1 with 32 bytes of data:

Reply from 216.90.90.1: bytes=32 time=45ms TTL=126
Reply from 216.90.90.1: bytes=32 time=51ms TTL=126
Reply from 216.90.90.1: bytes=32 time=44ms TTL=126
Reply from 216.90.90.1: bytes=32 time=43ms TTL=126

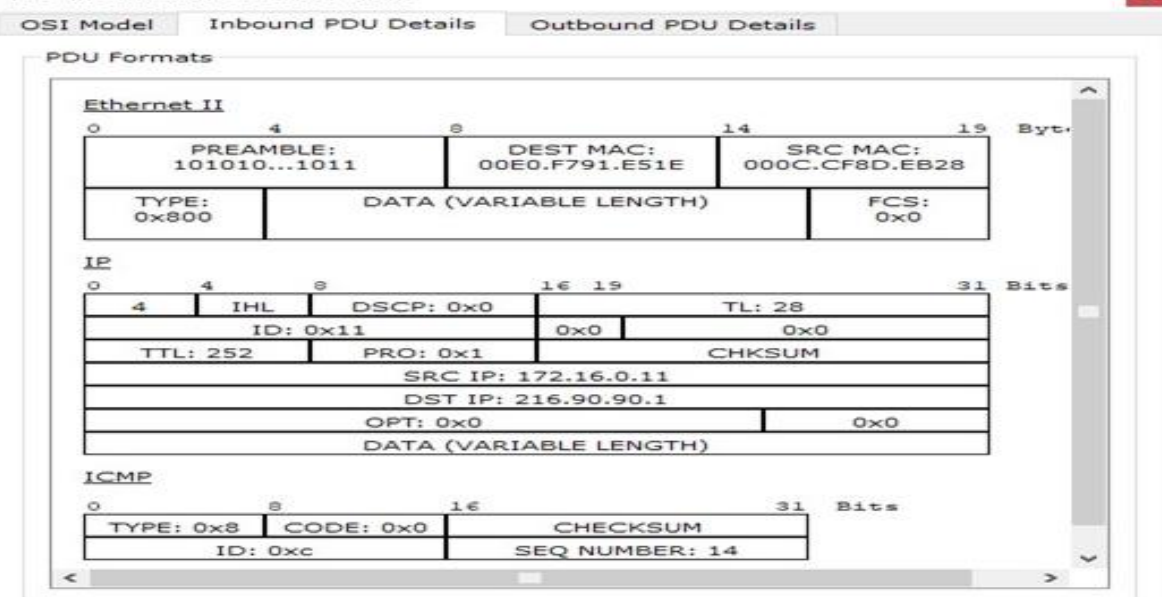
Ping statistics for 216.90.90.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 43ms, Maximum = 51ms, Average = 45ms

SERVER>

```



PDU Information at Device: Client



✓ Client vers PC1 ou 2

```
PC>ping 172.16.0.11

Pinging 172.16.0.11 with 32 bytes of data:

Reply from 172.16.0.11: bytes=32 time=47ms TTL=125
Reply from 172.16.0.11: bytes=32 time=57ms TTL=125
Reply from 172.16.0.11: bytes=32 time=46ms TTL=125
Reply from 172.16.0.11: bytes=32 time=44ms TTL=125

Ping statistics for 172.16.0.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 44ms, Maximum = 57ms, Average = 48ms

PC>
```

✓ De client vers www.entreprise.com

```

PC>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time=45ms TTL=126
Reply from 10.0.0.3: bytes=32 time=58ms TTL=126
Reply from 10.0.0.3: bytes=32 time=42ms TTL=126
Reply from 10.0.0.3: bytes=32 time=69ms TTL=126

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 42ms, Maximum = 69ms, Average = 53ms

PC>

```

4. Protection du réseau de l'entreprise

En entrée (ACL 101)

```

Private (config)#access-list 101 permit ip 172.16.0.0 0.0.0.255
10.0.0.0 0.0.0.255
` 216.90.80.0 0.0.0.255

```

En sortie (ACL 100)

```

Private (config)#access-list 100 permit tcp host 172.16.0.0
Private (config)#access-list 100 permit icmp host 10.0.0.3 host 172.16.0.0
Private (config)#access-list 100 permit icmp host 172.16.0.0 host 10.0.0.0 host
216.90.80.0

```

Activons les listes sur l'interface du routeur

```

Private (config)#access-list 101 in

```

```

Private (config)#access-list 100 in

```

Tableau des résultats après filtrage

Source	Destination	Résultat
PC1 ou 2	Client	Succès (Le trafic sort via l'ACL 101, revient via le mot-clé established de l'ACL 100).

	Client	Succès (Le serveur web répond aux requêtes du client externe sans interférence de private).
Client	PC1 ou 2	Échec (Bloqué en entrée par le routeur private via l'ACL 100 car ce n'est ni un echo-reply ni une connexion TCP established).
Client		Succès (Le trafic s'arrête à la DMZ, le réseau interne n'est pas sollicité).

4.5 Protection de la DMZ

Nous voulons restreindre l'accès à la DMZ depuis l'extérieur sur le routeur public via l'ACL étendue 111, appliquée en entrée sur son interface externe (côté Internet, ex: Fa0/1).

Logique de l'ACL 111

Autoriser le trafic HTTP (port 80) vers le serveur Web public (216.90.80.253).

Autoriser les réponses TCP (port 80 de l'Internet vers les requêtes de nos utilisateurs internes).

Permettre le protocole de routage RIP (via UDP port 520) à circuler entre les routeurs.

Autoriser les pings de type echo-reply (réponses) destinés au serveur web de l'entreprise.

Configuration sur public

```
public(config)# access-list 111 permit tcp any host 216.90.80.253 eq www
public(config)# access-list 111 permit tcp any eq www host 216.90.80.254
established
public(config)# access-list 111 permit udp any any eq rip
public(config)# access-list 111 permit icmp any host 216.90.80.253 echo-reply
```

Application sur l'interface externe :

```
public(config)# interface FastEthernet0/1  
public(config-if)# ip access-group 111 in
```

Réponse : C'est la règle implicite de fin d'ACL (deny ip any any) qui s'est appliquée. Comme le Client tente de faire une requête DNS (UDP port 53) directement vers l'adresse de la DMZ et que ce protocole n'est pas explicitement autorisé dans l'ACL 111, le paquet est immédiatement jeté (Dropped).

4.6 Conclusion

Le protocole FTP utilise deux ports distincts : le port 21 (commandes/contrôle) et le port 20 (données en mode actif). Pour autoriser ce service, il faudrait ajouter les instructions suivantes :

Sur le routeur public (dans l'ACL 111 en entrée) :

```
public(config)# access-list 111 permit tcp any host 216.90.80.253 eq 21  
public(config)# access-list 111 permit tcp any host 216.90.80.253 eq 20
```

Sur le routeur public (pour la translation NAT) : L'instruction NAT statique globale déjà configurée (ip nat inside source static 10.0.0.3 216.90.80.253) convertit l'ensemble des ports pour cette adresse IP, donc aucune modification NAT supplémentaire n'est requise.